B. Tech II nd year
C.S

University Roll No._____

GLA UNIVERSITY, MATHURA
B. Tech. 2nd Yr. (Even Sem.), Term II Examination, 2016-17

Subject: English for Professional Purposes-II | Subject Code: AHE 4083
Time: 45 Minutes | Max. Marks: 15

Note:

- **Answer all the questions from Section A, any 3 out of 4 from Section B, and any 2 out of 3 from Section C.**

Section A **0.5 x 6=3**

1. Differentiate between idiom and phrase.
2. Write career objective of your resume.
3. PPT is just an_____ to the presentation.
4. Give any 2 forms of pictorial representation of data.
5. Write one difference between CV and resume?
6. Identify any one role of cover letter in job application process.

Section B **2x3=6**

1. Write cover letter to apply for the post of GTE in BHEL, Haridwar considering you are a B. Tech graduate with GATE qualification.
2. Write technical review of any of these recently launched products:
   a. One plus 3t b. iphone 7
   c. Ignis d. Baleno RS
3. Imagine you are the president of students' union of your university, what are first 4 things you will do and why?
4. Make first four PPT slides on the topic **'Role of electronic gadgets in our lives'**, enumerating advantages, disadvantages, limitations etc.

Section C **3x2=6**

1. Write a letter of Enquiry to the 'IFFCO' to enquire about the availability and price of Urea, Phosphorus and DAP for use in your farm-land.
2. Describe the ISRO's '104 satellite launch in a single flight' story using following idioms, proverbs:
   Once bitten twice shy, actions speak louder than words, strike the iron when it is hot, make the most of.
3. The given graph is a representation of the number of runs scored in a match between Mumbai Indians and Chennai Super Kings on 5 May 2016, at Wankhade Stadium, Mumbai. The X-axis shows number of overs bowled, whereas the Y axis stands for number of runs scored in an over. Draw inferences and write the interpretation of the graph given overleaf:

SET A

University Roll No._____

B. Tech. 2nd Year (Even Semester), Term II Examination, 2017

Subject: English for Professional Purposes-II | Subject Code: AHE 4083

Time: 45 minutes | Max. Marks: 40

VERBAL APTITUDE

Sentences are given with blanks to be filled in with an appropriate option. Five alternatives are suggested for each question. Choose the correct alternative out of the five.

1. After resisting for two hours the rebellious leaders _____.
   A. gave off B. gave in C. gave out
   D. gave into E. gave on
2. A man of determination _____ all the difficulties of life.
   A. gets up B. gets on C. gets over
   D. gets by E. gets to
3. Sam could not _____ the insult so he lost his temper.
   A. put on with B. put through with
   C. put by with D. put in with
   E. put up with
4. Do not _____ my request without giving it a thought.
   A. turn in B. turn down C. turn at
   D. turn into E. turn off
5. The manager got _____ by emotions so he failed to take a sound decision.
   A. carried around B. carried in C. carried down
   D. carried away E. carried up
6. He has been appointed as the secretary _____ the minister.
   A. from B. to C. with
   D. for E. on
7. Cold wind came _____ the window!
   A. from B. across C. in
   D. through E. into
8. He has got a degree _____ Engineering.
   A. in B. at C. from
   D. on E. within
9. Why don't you come and meet me _____ the canteen.
   A. on B. under C. at
   D. from E. upon
10. The boy jumped into the river and swam it _____.
    A. on B. in
    C. at D. from
    E. across
11. Carlos is the only one of those students who _____ lived up to the potential described in the yearbook.
    A. has B. have C. will have
    D. A&C E. None of the Above
12. The International Club, as well as the Choral Society and the Rowing Club, _____ to submit a new constitution.
    A. need B. needs C. was needed
    D. is needed E. All of the Above
13. On June 30, 2017, the school _____ after the summer vacation.
    A. will open B. Opened C. opens
    D. has been opening E. will be opening
14. I cannot taste the grapes, it _____ sour.
    A. were tasting B. tastes C. taste
    D. tasting E. was tasting
15. Two children and one adult _____ in a fire last night.
    A. will die B. died C. have died
    D. are dying E. had died
16. _____ this is a waste of time.
    A. To put it in a nutshell
    B. To start the ball rolling
    C. To talk down
    D. To get the wrong end of the stick
    E. None of the Above
17. I'll be ready _____ after a long day at work.
    A. To pay over the odds
    B. To foot the bill
    C. To get a finger in every pie
    D. To crash out
    E. None of the Above
18. Come on! Hurry up! You're such a _____.
    A. Cold fish
    B. Slow coach
    C. As good as gold
    D. Odd ball
    E. None of the Above
19. I don't need a doctor. I just feel a bit _____.
    A. Jumping out of my skin
    B. As fit as fiddle
    C. Under the weather
    D. Shaking in my shoes
    E. None of the Above
20. The people in the office have been playing computer games all days since the boss fell ill. Well, it is rightly said _____.
    A. Don't count your chicken before they hatch.
    B. Never look a gift horse in the mouth.
    C. Don't put all your eggs in one basket.
    D. When the cat's away the mice will play
    E. None of the Above

Decide which sentence is the most preferable with respect to grammar/ meaning and usage, suitable for formal writing.

21. A. He will be likely to leave for the States last year to visit his wife.
    B. He will be likely leaving for the States last year to visit his wife.
    C. He was likely to leave for the States last year to visit his wife.
    D. He will likely to leave for the States last year to visit his wife.

E. He would likely to leave for the States last year to visit his wife.

Identify the part of speech of the underlined word:

22. She earns around forty thousand a year.
A. Adjective B. Noun
C. Pronoun D. Adverb
E. Preposition

23. This is a serious lacuna for a country that otherwise has a commitment to democracy and the rule of law.
A. Adverb B. Noun
C. Conjunction D. Preposition
E. Interjection

24. Few will argue that the rise in incomes that came with a more open economy has not translated into a higher quality of life for many Indians.
A. Adverb B. Adjective
C. Noun D. Preposition
E. Pronoun

25. Did you know Lucy was back in England?" "Is she - since when?"
A. Adjective B. Noun
C. Pronoun D. Verb
E. Adverb

26. I know very little about the whys and the wherefores of the situation.
A. Adverb B. Adjective
C. Verb D. Noun
E. Pronoun

Identify which part (A, B, C, D) of the sentence has an error.

27. The mother as well as (A)/ her children were (B)/ brought to the (C)/ police station for interrogation. (D)/ No Error. (E)

28. The result of the recognition (A)/ of these facts are seen (B)/ in the gradual improvement (C)/ of the diet of the poor. (D)/ No Error. (E)

29. I unfortunately missed (A)/ the last train which I usually catch (B)/ and have to stay back (C)/ at the waiting room of the station last night. (D)/ No Error. (E)

Do as Directed.

30. Has he submitted his assignment on time?
(Identify Tense and form)
A. Past Perfect in Interrogative form
B. Present Perfect in Interrogative form
C. Present Continuous Tense in Interrogative form
D. Present Indefinite in Interrogative form
E. Present Perfect Continuous in Interrogative form

Read the passage and answer the questions that follow:

Prior to the fall of the Union of Soviet Socialist Republics (USSR), Mikhail Gorbachev, seeing a country falling behind its Western rival and a people increasingly clamoring for change, addressed the growing internal unrest in the summer of 1987 by introducing a series of reforms known as perestroika (literally, restructuring). In *Perestroika: New Thinking for Our Country and the World*, Mikhail Gorbachev discussed his analysis of the problems facing the USSR and his plans to solve them.

Perhaps the most pressing and visible problem facing the USSR in the last 1980s came in the form of the country's consistently mediocre economic performance, despite its vast natural resource wealth and large labor force. Gorbachev flatly admitted that economic failures were increasing and current policies were failing to offer a sustainable remedy. Failing to take advantage of the numerous scientific and technological advancements available, the USSR relied on inefficient and outdated business models. As a result, Gorbachev said, "in the last fifteen years the national income growth rates had declined by more than a half and by the beginning of the eighties had fallen to a level close to economic stagnation." With business executives focused on using more resources (in order to employ more people) instead of becoming more efficient, the country produced poor quality products unable to compete in a global economy. Further, this inefficiency led to shortages: "the Soviet Union, the world's biggest producer of steel, raw materials, fuel and energy, has shortfalls in them due to wasteful or inefficient use."

The decrepit economy engendered social unrest and woe that only compounded economic difficulties and societal misery. Gorbachev wrote of "a gradual erosion of the ideological and moral values of our people" and noted the considerable growth in "alcoholism, drug addiction and crime." Accentuating these difficulties, the Communist government often ignored the needs of the average citizen, causing distrust and resentment. Perhaps the most destructive element of the social unraveling and inadequate government response was the mediocre education system. Gorbachev said, "Creative thinking was driven out from the social sciences, and superfluous and voluntarist assessments and judgments were declared indisputable truths."

Although Gorbachev also opined about the growing public disbelief in the content of the immense government propaganda campaigns, the extent to which economic underdevelopment and social deviance gripped Soviet culture made the collapse of the USSR virtually inevitable in the minds of many observers. When combined with *glasnost* (literally, openness), Gorbachev's plan that allowed greater transparency, *perestroika* actually served to hasten the collapse of the USSR. Contrary to its purpose, *perestroika* ensured that the fall of the USSR would occur sooner rather than later. Only a few years after Gorbachev implemented changes that would have been unthinkable and antithetical to the philosophy of previous leaders like Lenin, Stalin, and Khrushchev, the USSR fell.

31. Which of the following best describes the primary objective of the passage?

A. Argue that the implementation of perestroika caused the fall of the Soviet Union
B. Explain perestroika along with its roots and consequences
C. Analyze the pros and cons of Mikhail Gorbachev's decision to implement perestroika
D. Explain the short-falls of a communist system and offer remedies
E. Discuss the role of Mikhail Gorbachev in propelling the USSR towards ceasing to exist

32. The passage implies that which of the following was most true of the Soviet economy prior to perestroika:
A. Suffered from underperformance due to excessive government regulation and micro-management
B. Failed to meet its potential as a result of corruption and bureaucratic overhead
C. Lacked adequate natural resources to grow efficiently, regardless of business management
D. Focused on achieving high-employment rather than export-capable products
E. With declining growth and stagnation, USSR stood in the worst shape ever in its history

33. Based upon the passage, the author would likely agree most with which of the following characterizations of the impact of the USSR's troubled economy during the days leading up to perestroika?
A. Cause for renewed determination in communist philosophy
B. Reason that natives looked increasingly to the West and capitalism
C. Source of frustration and discomfort among citizens that fueled social friction
D. Justification for the USSR's neglect of the needs of many citizens
E. Primary cause of the USSR's poor educational system

34. According to the passage, which of the following best describes the relationship between perestroika and the fall of the USSR?
A. Perestroika mildly delayed the fall of the USSR, although the decline of the Soviet Republic was inevitable
B. Perestroika hastened the decline of the USSR
C. Perestroika enabled the USSR to pursue much needed restructuring
D. Perestroika softened the impact from the collapse of the USSR
E. Perestroika had little relationship to the decline of the USSR, which was inevitable anyway

35. In the context of the passage, the author most likely uses the word "unthinkable" (in the last sentence) to help convey which of the following points about the changes Gorbachev implemented in perestroika?
A. They would have never crossed the mind of Lenin as being conceptually possible, let alone desirable or feasible
B. They would have been difficult for the mind of Lenin to comprehend intelligibly
C. They would have been seen by Lenin as undesirable and poor choices
D. They would have been seen by Lenin as incomprehensible yet appealing
E. They would have been considered highly desirable

36. Gorbachev offers all of the following as evidence of the need for perestroika EXCEPT:
A. Shortages in natural resources due to inefficiency
B. Declines in economic output and growth
C. Slides in moral values of citizens
D. Erosion of new and ingenious thinking
E. Frustrations with the results of past reform efforts

37. Which of the following words best describes the passage's tone?
A. Primarily Analytical
B. Highly Critical
C. Frustrated
D. Not Objective
E. Deeply Introspective

Read the Passage and answer the questions that follow:

One of my favorite vacation places is Mexico. I really like the weather there because it never gets cold. The people are very nice too. They never laugh at my bad Spanish. The food is really good. Mexico City is a very interesting place to visit. It has some great museums and lots of fascinating old buildings. The hotels are too expensive to stay but there are more affordable options. For example, you can stay at one of the beach resorts like Acapulco. If you are planning to visit Mexico, you should definitely see the Mayan temples near Merida.

38. Which one is correct according to the passage:
A. Sam likes Warm weather.
B. Sam doesn't like warm weather at all
C. Sam hates warm weather
D. Sam likes cold weather
E. Sam has no preference about the weather.

39. How is his Spanish:
A. Excellent
B. Good
C. Bad
D. Very bad
E. He doesn't know any Spanish

40. Which one is true about Mexico according to the passage:
A. There's a lot to see and do in Mexico
B. There aren't a lot of beautiful places in Mexico
C. Mexico is a dirty place
D. Tourists never come to Mexico
E. You have no option but to spend a lot of money if you wish to stay at Mexico.

E. He would likely to leave for the States last year to visit his wife.

Identify the part of speech of the underlined word:

22. She earns around forty thousand a year.
A. Adjective B. Noun
C. Pronoun D. Adverb
E. Preposition

23. This is a serious lacuna for a country that otherwise has a commitment to democracy and the rule of law.
A. Adverb B. Noun
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24. Few will argue that the rise in incomes that came with a more open economy has not translated into a higher quality of life for many Indians.
A. Adverb B. Adjective
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25. Did you know Lucy was back in England?" "Is she - since when?"
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31. Which of the following best describes the primary objective of the passage?

SET B University Roll No._____

B. Tech. 2nd Year (Even Semester), Term II Examination, 2017

Subject: English for Professional Purposes–II Subject Code: AHE 4083

Time: 45 minutes Max. Marks: 40

VERBAL APTITUDE

Sentences are given with blanks to be filled in with an appropriate option. Five alternatives are suggested for each question. Choose the correct alternative out of the five.

1. The employee_____ his intentions without any fear.
   A. gave out B. gave in C. gave to
   D. gave around E. gave into
2. Though he is weak at studies yet he managed to _____ the examination.
   A. get in B. get along C. get through
   D. get with E. get over
3. Do not _____ till tomorrow what you can do today.
   A. put up B. put with C. put on
   D. put of E. put off
4. Renu was relying on her boyfriend but he_____ to be a cheat.
   A. turned up B. turned of C. turned over
   D. turned out E. turned off
5. _____ your work with the same pace so that you can finish it in time.
   A. Carry through B. Carry on
   C. Carry along D. Carry in
   E. Carry off
6. The newly appointed deputy Ambassador _____ the UN has raised the issue of cross border terrorism.
   A. from B. to C. with
   D. for E. on
7. Germany's MEG-LEV train clocked 500kms/hr. while passing _____ the newly build Berlin-Bonn tunnel.
   A. from B. across C. in
   D. through E. into
8. Instead of studying Economics, he wished to get a degree _____ History.
   A. in B. at C. from
   D. on E. within
9. The person whom you met _____ the party was my brother.
   A. on B. under C. at
   D. from E. upon
10. My friend lives just _____ the street.
   A. upon B. into C. at
   D. from E. across
11. Not only the students but also their instructor _____ been called to the principal's office.
   A. has B. have C. was
   D. are E. is
12. Most of the milk _____ gone bad. Six gallons of milk _____ still in the refrigerator.
   A. has, are B. have, is C. are, is
   D. is, are E. was, is
13. For dinner last night we _____ fish and chips.
   A. will have B. had
   C. are having D. have had
   E. had had
14. Please be quiet! The baby _____
   A. sleeps B. is sleeping
   C. has slept D. sleep
   E. will sleep
15. I _____ to the cinema tonight: I've got a lot of work to finish.
   A. will not go B. 'm going
   C. will go D. 'm not going
   E. went
16. After 2 months of gruelling schedule, I _____ to go for a vacation in the Himalayas.
   A. want B. wanting C. am wanting
   D. have been wanting
   E. has been wanting
17. She is very direct and always _____.
   A. Speaks her mind
   B. Beat around the bush
   C. Talk shop
   D. Talk behind her back
   E. None of the Above

18. She is _____ above the rest of the girls.
A. Top notch
B. The cat's whiskers
C. Head and shoulders
D. Dog's breakfast
E. None of the Above

19. He is quite out of touch with reality. He has his head _____.
A. In the clouds
B. Over the top
C. Round the corner
D. Middle of the road
E. None of the Above

20. Joy is always criticizing people who are selfish yet he is terribly selfish himself. Well, that reminds me of _____.
A. People who live in glass houses should not throw stones
B. Absence makes the heart grow fonder
C. A bird in the hand is worth two in the bush
D. There is no smoke without fire.
E. None of the Above

21. Don't _____ with Shreya. That will do you no good.
A. Be spick and span
B. Be a feather in the cap
C. Lock horns
D. A cock and bull story
E. None of the Above

Decide which sentence is the most preferable with respect to grammar/ meaning and usage, suitable for formal writing.

22.
A. Not seldom he comes late to the school.
B. Does he come late to the school?
C. Is he come late to the school?
D. Do he come late to the school?
E. Does he comes late to the school?

Identify the part of speech of the underlined word:

23. She looks best in bright, vibrant colours, <u>like</u> red and pink.
A. Adverb B. Adjective C. Verb
D. Conjunction E. Preposition

24. This is the standard model, <u>then</u> there's the deluxe version which costs more.
A. Adverb B. Adjective C. Verb
D. Noun E. Preposition

25. The letter showed clearly <u>what</u> they were planning.
A. Adverb B. Adjective C. Noun
D. Preposition E. Pronoun

26. Theirs was the most <u>storied</u> romance in Hollywood.
A. Adjective B. Noun C. Pronoun
D. Verb E. Preposition

27. He's lived in Mathura for 40 years, or <u>thereabouts.</u>
A. Verb B. Adverb
C. Adjective D. Noun
E. Conjunction

Identify which part (A, B, C, D, E) of the sentence has an error.

28. He like the other (A)/ members of his (B)/family were left shelter less (C)/ as a result of (D)/ flood in the town (E).

29. Intelligence, as well as (A)/ knowledge of the subject (B)/ are required to grasp (C)/ the meaning of the book. (D)/ No Error. (E)

30. After you will return (A)/ from Agra (B)/ I will come to meet you (C)/ at your Park Street house. (D) / No Error. (E)

Read the passage and answer the questions that follow:

On August 22, 1939, Adolf Hitler summoned his top military generals to Obersalzberg, where he delivered a speech explaining his plans for war, first with Poland, then with the rest of Europe. Despite resistance from those both inside and outside Germany, Hitler felt exceedingly confident that he could defy the will of the international community and conquer vast amounts of land. In his speech at Obersalzberg, he laid out numerous factors he believed would contribute to the success of his war plans.

Chief among Hitler's sources of confidence in Germany's brazen war plans was German military quickness. Hitler said, "Our strength lies in our quickness." On the advice of Colonel-General von Brauchitsch, Hitler believed Poland could be captured in a few weeks, an astonishingly short amount of time given the recent history of trench warfare and the long history of protracted European military engagements that resulted in minimal land gains and high casualty counts.

Hitler's confidence in the ability of the German military to inflict considerable brutality further strengthened his determination to pursue an exceedingly ambitious plan of territorial aggrandizement. He said, "I shall shoot everyone who utters one word of criticism" and noted that "the goal to be obtained in the war is not that of reaching certain lines but of physically demolishing the opponent."In this vein, Hitler ordered his military to "be hard, be without mercy, [and] act more quickly and brutally than others...for it scares the others off." Hitler believed that enemies, not used to this type of brutality, would surrender quickly.

In addition to speed and brutality, Hitler believed that, in the end, history would overlook his inhumane conduct. To support this view, which turned out to be anything but prescient, Hitler invoked a pollyannaish view of Asian leader Genghis Khan. In Hitler's eyes, Khan "sent millions of women and children into death knowingly and with a light heart," yet "history sees in him only the great founder of States."

Although Hitler brimmed with confidence and experienced initial yet widely-expected success in Poland and then in Denmark, he overlooked important considerations. In many ways, Hitler made the same mistake Napoleon Bonaparte made years earlier. Hitler believed he could advance further and conquer Britain, yet, like Napoleon, Hitler did not adequately foresee the insurmountable barrier posed by Britain's island status. Despite the damage inflicted at the hands of the German Luftwaffe during the Battle of Britain (1940), British forces eventually won this important battle. Nevertheless, Hitler pressed on and, in an even more fateful decision that carried echoes of a Napoleonic tactical misstep, invaded the USSR where his forces suffered the decisive defeat of World War II at Stalingrad in 1943. In the end, Hitler's reputation in history proved to be as brutal and decisive as the battle plans and philosophy he announced at Obersalzberg.

31. According to the passage, Hitler's confidence in his military strategy stemmed from its:
A. Surprise invasions
B. Emphasis on unconventional warfare
C. Reliance on air supremacy
D. Swift brutality
E. Napoleonic overtones

32. The author of the passage is primarily concerned with explaining:
A. The logistics of Hitler's war strategy and the mechanics of its failure
B. The philosophy of Hitler's war strategy and the world's reaction
C. Why Hitler believed his war plans would succeed and why they eventually failed
D. Hitler's plans and their failure with an eye to pre-1900 history
E. Explaining the source of Hitler's brutality and the reasons for its failure

33. Which of the following best characterizes the author's view of the relationship between Hitler and Napolean?
A. Governed with similar styles
B. Fought military conflicts with similar ideologies

C. In general, shared a legacy as overly ambitious leaders
D. At a high-level, some similarities in military missteps existed
E. Both suffered final defeats by impetuously charging east

34. According to the passage, what best describes the author's understanding of why Hitler's military campaign eventually failed?
A. Failed to demoralize opponents
B. Overlooked important tactical and geographic considerations
C. Underestimated international resolve
D. Fell behind technological advancements of European enemies
E. Failed to consolidate initial military and land gains

35. According to the passage, Hitler's confidence in the war plans announced at Obersalzberg stemmed from all of the following EXCEPT:
A. The speed of the German military
B. The brutality of the German military
C. The plan to stifle dissent
D. The belief that instilling fear weakened enemies
E. The history of overlooking European military brutality

36. According to the passage, why did Hitler believe he could conquer Poland in a few weeks?
A. The inaction of European neighbors
B. The example of Napoleon
C. The philosophy of Genghis Khan
D. The counsel of a military general
E. The small size of Poland

37. According to the passage, which of the following represents the chronological unfolding of events?
A. Generals summoned to Obersalzberg; Invaded Poland; Invaded Denmark; the Battle of Britain; Battle at Stalingrad
B. Generals summoned to Obersalzberg; Invaded Denmark; Invaded Poland; the Battle of Britain; Battle at Stalingrad
C. Generals summoned to Obersalzberg; Invaded Denmark; Invaded Poland; Battle at Stalingrad; the Battle of Britain
D. Generals summoned to Obersalzberg; Invaded Poland; Invaded Denmark; Battle at Stalingrad; the Battle of Britain
E. Generals summoned to Obersalzberg; the Battle of Britain; Invaded Poland; Invaded Denmark; Battle at Stalingrad

Read the Passage and answer the questions that follow:

Last summer, we decided to spend our vacation at the beach because the weather was very hot in the mountains. The travel agent said that traveling by bus was the cheapest way, but we went by plane because it was faster. We wanted to have more time to spend at the beach. The weather was beautiful and we had a great time.

38. We decided to go to the beach because...
A. it was cheaper than going to the mountains
B. the travel agent said that it was the cheapest
C. of the hot weather in the mountains
D. we wanted to spend time at the beach
E. None of the above

39. The bus was the ---- way to travel.
A. Best
B. Easiest
C. Cheapest
D. Slowest
E. Fastest

40. What were the options available for travelling:
A. Bus and Plane
B. Only Bus
C. Only Plane
D. Train and Bus
E. Train and Plane

SET C

University Roll No._____

B. Tech. 2nd Year (Even Semester), Term II Examination, 2017

Subject: English for Professional Purposes–II | Subject Code: AHE 4083

Time: 45 minutes | Max. Marks: 40

VERBAL APTITUDE

Sentences are given with blanks to be filled in with an appropriate option. Five alternatives are suggested for each question. Choose the correct alternative out of the five.

1. Peter could not stand against opposition so he had to _____.
   A. give in B. give into C. give off D. give over E. give of
2. If you interfere in other's affairs you will _____ trouble.
   A. get in B. get on C. get into D. get through E. get over
3. The teacher has _____ his point of view-accept or reject it.
   A. put ahead B. put forth C. put on D. put of E. put at
4. Rohan generally _____ at odd hours.
   A. turns in B. turns at C. turns over D. turns up E. turns by
5. You must _____ the orders given to you.
   A. carry off B. carry around C. carry of D. carry through E. carry out
6. The Prime Minister's special envoy _____ the North East has announced a new developmental package for the region.
   A. from B. to C. with D. for E. on
7. He peeped _____ the window to find the contents of the room.
   A. from B. across C. in D. through E. into
8. Despite being a doctor by profession Roget has a lot of interest _____ the origin of words.
   A. in B. at C. from D. on E. within
9. _____ the cinema, I felt a little bit uneasy.
   A. on B. under C. at D. from E. upon
10. You need to walk _____ the bridge to get to Paharganj.
    A. on B. in C. at D. from E. across
11. Each and every student and instructor in this building _____ for a new facility by next year.
    A. hope B. hopes C. hoped D. is hoping E. was hoping
12. They would have won if they _____ better.
    A. will play B. had played C. have play D. have played E. played
13. Raj usually _____ exercises daily in the evening.
    A. has taken B. takes C. do D. is doing E. does
14. When she came into the office, I _____ to the manager.
    A. talked B. was talking C. have talked D. have been talking E. had talked
15. The Prime Minister _____ America next week.
    A. will be visiting B. will visit C. is to visit D. will have been visiting E. has been visiting
16. Suddenly, from the balcony, a little boy stood up and _____, "Daddy, I think you're wonderful!"
    A. shouts B. was shouting C. had shouted D. shouted E. None the Above
17. We had to _____ for those tickets.
    A. Pay through the nose
    B. Got a finger in every pie
    C. Get it off our head
    D. Make a headway
    E. None of the Above
18. It's midnight. Time _____.
    A. To down and out
    B. To Fly off the handle
    C. To hit the sack
    D. To be out of sorts
    E. None of the Above
19. Safeena seems to be _____ these days.
    A. On cloud nine
    B. Over the moon
    C. On the top of the world
    D. All of the Above
    E. None of the Above
20. I got him enrolled in the best coaching institute in Mathura but he did not attend the classes regularly. As they say _____.
    A. You can lead a horse to the water but you can't make it drink
    B. Many heads make light work
    C. One swallow does not make a summer
    D. Too many cooks spoil the broth
    E. None of the Above
21. It's a secret. Make sure to _____.
    A. Keep it under wraps
    B. Bite off more than you can chew
    C. Burn the midnight oil
    D. Cross the bridge when you come to it
    E. None of the Above

Decide which sentence is the most preferable with respect to grammar/ meaning and usage, suitable for formal writing.

22. A. The rest of the students is still working in the class.

B. The rest of the students are still working in the class.
C. The rest of the students still works in the class.
D. The rest of the students are still work in the class.
E. The rest of the students is still work in the class.

23. A. The soldiers along with the Commander was court- martialled for defying the orders of Army High-Command.
B. The soldiers along with the Commander is court-martialled for defying the order of Army High-Command.
C. The soldiers along with the Commander were court-martialled for defying the order of Army High-Command.
D. The soldiers along with the Commander will court-martialled for defying the order of Army High-Command.
E. The soldiers along with the Commander has been court-martialled for defying the order of Army High-Command.

Identify the part of speech of the underlined word.

24. The cost of car insurance is up, but not very much.
A. Verb B. Adverb C. Adjective
D. Noun E. Preposition

25. The train suddenly started moving as I was getting on.
A. Adverb B. Adjective C. Conjunction
D. Noun E. Pronoun

26. These shoes have leather uppers and synthetic soles.
A. Adverb B. Adjective C. Verb
D. Noun E. Pronoun

27. Do not bank upon the bank that is close to your residence. Their mode of operation creates suspicion.
A. Verb and Verb B. Noun and Verb
C. Adjective and Verb D. Noun and Adjective
E. Noun and Verb

28. Let us never forget those who gave their lives for their country.
A. Adjective B. Adverb C. Verb
D. Noun E. Pronoun

Identify which part (A, B, C, D, E) of the sentence has an error.

29. The number of amendments (A)/ to our constitution have been (B)/ very large during (C)/ the last fifty (D)/ years of independence (E).

30. Either she or you (A)/ is to blame for (B)/ the management of (C)/ the domestic affairs. (D)/ No Error. (E)

Read the passage and answer the questions that follow:

Marketing executives in television work with a relatively stable advertising medium. In many ways, the television ads aired today are similar to those aired two decades ago. Most television ads still feature actors, still run 30 or 60 seconds, and still show a product. However, the differing dynamics of the Internet pose unique challenges to advertisers, forcing them to adapt their practices and techniques on a regular basis.

In the early days of Internet marketing, online advertisers employed banner and pop-up ads to attract customers. These techniques reached large audiences, generated many sales leads, and came at a low cost. However, a small number of Internet users began to consider these advertising techniques intrusive and annoying. Yet because marketing strategies relying heavily on banners and pop-ups produced results, companies invested growing amounts of money into purchasing these ad types in hopes of capturing market share in the burgeoning online economy. As consumers became more sophisticated, frustration with these online advertising techniques grew. Independent programmers began to develop tools that blocked banner and pop-up ads. The popularity of these tools exploded when the search engine Google, at the time an increasingly popular website fighting to solidify its place on the Internet with giants Microsoft and Yahoo, offered free software enabling users to block pop-up ads. The backlash against banner ads grew as new web browsers provided users the ability to block image-based ads such as banner ads. Although banner and pop-up ads still exist, they are far less prominent than during the early days of the Internet.

A major development in online marketing came with the introduction of pay-per-click ads. Unlike banner or pop-up ads, which originally required companies to pay every time a website visitor saw an ad, pay-per-click ads allowed companies to pay only when an interested potential customer clicked on an ad. More importantly, however, these ads circumvented the pop-up and banner blockers. As a result of these advantages and the incredible growth in the use of search engines, which provide excellent venues for pay-per-click advertising, companies began turning to pay-per-click marketing in droves. However, as with the banner and pop-up ads that preceded them, pay-per-click ads came with their drawbacks. When companies began pouring billions of dollars into this emerging medium, online advertising specialists started to notice the presence of what would later be called click fraud: representatives of a company with no interest in the product advertised by a competitor click on the competitor's ads simply to increase the marketing cost of the competitor. Click fraud grew so rapidly that marketers sought to diversify their online positions away from pay-per-click marketing through new mediums.

Although pay-per-click advertising remains a common and effective advertising tool, marketers adapted yet again to the changing dynamics of the Internet by adopting new techniques such as pay-per-performance advertising, search engine optimization, and affiliate marketing. As the pace of the Internet's evolution increases, it seems all the more likely that advertising

successfully on the Internet will require a strategy that shuns constancy and embraces change.

31. The author implies what about the future of pay-per-performance advertising?
A. Although it improves on pay-per-click advertising, it is still vulnerable to click fraud
B. It will one day become extinct as Internet users discover drawbacks with it
C. Internet users will develop free software to block its effectiveness
D. It will eventually become less popular with advertisers as the Internet evolves and drawbacks emerge
E. It will not face drawbacks due to its differing approach to online marketing

32. Which of the following most accurately states the main idea of the passage?
A. Although pay-per-click advertising remains a wide-spread and effective online advertising medium, its popularity is likely to diminish as the Internet evolves
B. Internet advertising is not well received by Internet users, causing independent programmers to subvert advertisers.
C. Unlike the television, the Internet has experienced dramatic changes in short periods of time.
D. Unlike the television, the Internet has evolved rapidly, forcing online marketers to develop new advertising strategies and mediums.
E. The pace of the Internet's evolution is increasing and will only increase in the future.

33. According to the passage, which of the following best describes the current status of pop-up ads?
A. Widely used
B. Less popular now than at earlier times
C. A frequent target of click fraud
D. Non-existent due to pop-up blockers
E. Increasingly popular due to search engines

34. According to the passage, which of the following best describes the practice of click fraud?
A. Clicking on the banner advertisements of rival companies
B. Using software to block advertisements
C. Utilizing search engine optimization to visit the pages of competitors
D. Fraudulently purchasing products online
E. Clicking on the pay-per-click ads of competitors

35. According to the passage, the largest point at which the television and Internet differ as an advertising medium is:
A. The type of individual each medium reaches
B. Whether the medium is interactive
C. The pace at which the medium evolves
D. The cost of advertising with each medium
E. Whether each medium contains drawbacks

36. The passage implies that which of the following attributes will be the most important for future success in online advertising:
A. Flexible creativity
B. Disciplined patience
C. Uniform approach
D. Ruthless tenacity
E. Eclectic approach

37. Which of the following words best describes the author's tone in the passage?
A. Critical
B. Analytical
C. Frustrated
D. Biased
E. Surprised

Read the passage and answer the questions that follow:

Dear Daniel,

If you'd like to improve your English, one thing you can do is to build up your vocabulary. In order to do this, you have to practice a lot. I'm afraid there is not a short cut for this. But there are many effective ways. First of all, you should read a lot. You can read short stories, comics, newspapers, magazines etc. depending on your English level. As you read, you can try to guess the meanings of unknown words from the context of the sentence. If you cannot, then look up the definition in a dictionary. English to English dictionary should be your first choice and keep the translator as last. Another way of building vocabulary is to watch movies in English which will also help with pronunciation. Try to be involved with the language as much as you can. Keep a vocabulary notebook and write down a few words each day. Keep in mind that you have to revise them regularly. In my opinion, to communicate with people all you need is words. Even if you just say the words one after another, people most likely will understand what you mean regardless of the order of the words and grammar.

38. What is the main point of this e-mail?
A. What can we do to improve our English
B. Difficulties of learning English
C. Problems of learning a second language alone
D. What's the best way of learning English
E. It is not about any particular topic

39. Which of the following is incorrect?
A. We should use a translator for all the unknown words.
B. We should read a lot.
C. We should take notes when we learn new words.
D. We should practice as much as we can.
E. Revise the words learnt regularly.

40. What is the most important thing to communicate with people according to the writer?
A. Grammar
B. Vocabulary
C. Order of the words used
D. Advanced English course
E. Perfect usage of vocabulary

Paper Code- AHE4081

SET-C

GLA University

Soft Skills

Second Mid Term Examination (2016-17)

Course- B.Tech

Year- II

Branch- All Branches

Instructions :

1. There are 2 sections in the paper.
2. Attempt all questions. The maximum marks for the test are 20.
3. The total time allocated is 60 minutes.
4. There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.
5. The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.
6. The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.
7. Only black& blue pen should be used to fill forms.
8. Each branch is identified with a specific subject code which should be written in subject column of OMR sheet
9. Write your University roll number in the roll no column of OMR sheet.

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:.....

Signature of Invigilator-.....

S.N	Branch	Code
1	CS	1500
2	CE	1000
3	ECE	1300
4	EN	1400
5	EE	1100
6	ME	1200
7	MCA	8840

All The Best

LIBGLA

Section-1 (Reasoning Ability)- No of Questions-20, marks-10

Q(1/4)- A large cube is painted blue on all six faces and then cut into 64 smaller identical cubes.It was found that amongst the smaller cubes, there were 8 which had no face painted at all.

Q1. How many small cubes have exactly one face painted blue?
A.18 B.8 C.28 D.24

Q2. How many small cubes have at least one face painted blue?
A.65 B.56 C.57 D.58

Q3. How many small cubes have exactly three faces painted blue?
A.18 B.8 C.28 D.38

Q4. How many small cubes have at least two faces painted blue?
A.12 B.16 C.30 D.32

Q(5/6)-In a cube two opposite faces are Blue and Red.. Green is adjacent to Orange and Black.One remaining face is brown.

Q5.Which of the statement(s) is/ are definitely true?
I. Green and Brown are adjacent faces.
II.Black and Orange are adjacent faces.
III. Black face is between Blue and Red.

A.I and II only B.II and III only C.III only D.II Only

Q6. Which face is opposite to Brown face?
A.Orange B.Red C.Green D.Blue

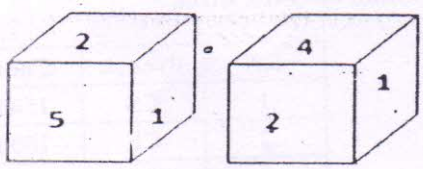
Q(7/9)- A cube of side 10 cm is painted with a 2 cm wide Green strip along all the side on all faces and the remaining space enclosed by this 2 cm strip is painted Red. The cube is cut into 125 smaller cubes of equal size. Now answer the following question based on the above information.

Q7. How many cubes have three Green faces?
A.0 B.4 C.6 D.8

Q8. How many cubes have one face Red and adjacent face Green?
A.0 B.4 C.6 D.8

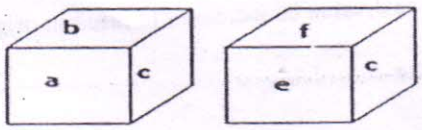
Q9. How many cubes have at least one face painted?
A.25 B.50 C.98 D.100

Q10. Two positions of dice are shown below. What number will appear on the face opposite to the face having '5'?



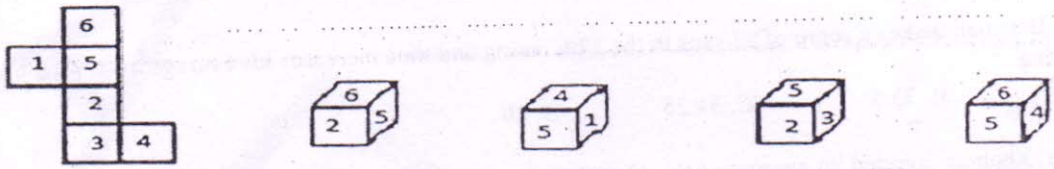
A. 2 B. 4 C. 1 D. 6

Q11. Which digit will appear on the face opposite to the face with alphabet 'd'?



A. a B. g C. c D. f

Q12.The figure given on the left hand side, in this problem, is folded to form a cube .Choose the well-formed cube from the given alternatives which is similar to the given design(X).



(X) (a) (b) (c) (d)

Q13. A child is looking for his father. He went 90 meters in the east before turning to his right. He went 20 meters before turning to its right again to look for his father at his uncle's place he went 30 meters from that point, But his father was not there, From there, he went 100 meters to his north before meeting his father in a street. How far did the son meet his father from starting point?

A.80 meters B.90 meters C.100 meters D.110 meters

Q14. Town D is towards East of town F .Town B is towards North of town D.Town H is towards South of town B. Now, towards which direction town H is from town F?

A.Data inadequate B.North-East C.East D.South-East

Q15. A is the sister of B and C. D is the father of B and G is the mother of A. Which of the following statements is definitelynot true?

A.G is Mother of B B.D is husband of G C.B is father of C D.A is daughter of D

Q16. Pointing to a gentleman, Shalini said, "His only brother is the father of my son's father." How is the gentleman related to Shalini ?.

A.Grandfather B.Uncle C.Brother D.Father

Q17. In a certain code, '37' means 'which class' and '583' means 'caste and class'. What is the code for 'caste' ?

A.7 B.3 C.Either 5 or 3 D.Either 5 or 8

Q18. In a certain code, MENTION is written as LNEITNO. How is PATTERN written in that code?

A. ATAETNR B. OTAETNR C. OTAESNR D. STAETNR

Q19. In a row of trees, one tree is fifth from either end of the row. How many trees are there in the row?

A. 8 B. 9 C. 10 D. 11

Q20. If Atul finds that he is twelfth from the right in a line of boys and fourth from the left, how many boys should be added to the line such that there are 28 boys in the line?

A. 12 B. 13 C. 14 D. 20

Section-2 (Quantitative Aptitude)- No of Questions-20, marks-10

Q 21. If 7X5462 is divisible by 9. Find X.
A. 3 B. 5 C. 4 D. 8

Q 22. A number is decreased by 6.66%. By what percent should the new value be increased to result in original value again?
A. 6.25% B. 7.14% C. 5.25% D. None of these

Q 23. A trader's marked price is 20% above cost price. He allows his customers 10% discount on the marked price. What profit does he make?
A. 10% B. 12% C. 15% D. 8%

Q 24. Calculate the Compound interest on Rs.-8800, if the rate of interest for 1st year is 12.5% and for 2nd year is 9.09 %.
A. 2300 B. 2000 C. 2200 D. 1600

Q 25. A Batsman makes a score of 87 runs in the 17th inning and thus increases his average by 3. Find his average after 17th inning.
A. 33.25 B. 33.5 C. 34.25 D. 39

Q 26. Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?
A. Rs. 6400 B. Rs. 7200 C. Rs. 6500 D. Rs. 7500

Q 27. A and B enter into partnership with investments of Rs. 54000 and Rs. 81000 respectively. A stayed for the entire year. If at the end of the year the profit was distributed equally, then for how many months less was B's investment there in the business?
A. 2 B. 4 C. 5 D. 6

Q 28. The speed of three cars is in the ratio of 5 : 3 : 2. What is the ratio among the times taken by these cars to travel the same distance?
A. 2 : 3 : 5 B. 5 : 2 : 6 C. 6 : 10 : 15 D. None of these

Q 29. The cost of a precious stone varies directly as the square root of its weight. A stone broke into 3 pieces whose weights are in the ratio 1 : 4 : 4. As a result, its value went up by Rs. 12000. Find its initial value.
A. 6,000 B. 18,000 C. 30,000 D. None of these

Q 30. How many kilograms of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per Kg so that there may be a gain of 10 % by selling the mixture at Rs. 9.24 per Kg ?
A. 36 Kg B. 42 Kg C. 54 Kg D. 63 Kg

Q 31. How many litres of water should be added to a 30 litre mixture of milk and water containing milk and water in the ratio of 7 : 3 such that the resultant mixture has 40% water in it?
A. 7 litres B. 10 litres C. 5 litres D. None of these

Q 32. From a cask of milk containing 30 litres, 6 litres are drawn out and is replaced with water. If the same process is repeated two more times, how many litres of milk will be left in the cask?
A. 0.512 litres B. 12 litres C. 14.38 litres D. 15.36 litres

Q 33. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 20%. Find the percentage of water in the mixture.
A. 25% B. 20% C. 6.66% D. 16.66%

Q 34. Find the sum of all the multiples of 6 between 200 and 1100.
A. 96, 750 B. 95, 760 C. 97, 560 D. 97, 650

Q 35. The sum of the first 13 terms of an arithmetic progression equal the sum of its first 27 terms. Find the sum of its first 40 terms.

A. 0 B. –1 C. 1 D. cannot be determined

Q 36. An equilateral triangle is formed by joining the mid points of the sides of another equilateral triangle. A third equilateral triangle is formed by joining the mid points of newly formed triangle and this process is continued infinitely. If area of outermost triangle is 120 cm^2, find the sum of the areas of all the triangles.

A. 90 cm^2 B. 120 cm^2 C. 160 cm^2 D. 200 cm^2

Q 37. Find the summation of the following series:

$$\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16}+\frac{5}{32}+....?$$

A. 5/2 B. 5 C. 10 D. None of these

Q 38. A fruit vendor has some Apples. When he packs apples in the packets of 3, 4 or 5 he is always left with 1 apple but when 7 apples are packed in a packet nothing is left. Find the minimum number of apples the fruit vendor possess.

A. 361 B. 301 C. 360 D. 421

Q 39. What is the least multiple of 9, which when divided by 7 and 11 gives remainder of 4 and 8 respectively?

A. 459 B. 474 C. 462 D. 477

Q 40. The least number which can be added to 532 so that it is completely divisible by 39 is?

A. 25 B. 39 C. 53 D. 14

LIFE@GLA

II Mid Term Examination, 2016-17

Course: B.Tech.(CSE) **Year: II** **Semester: IV**

Operating Systems (CSE4001)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. What are three requirements that a solution to the critical section problem must satisfy?

II. If the current value of counter = 4, what are its possible values if the producer and consumer processes run concurrently?

III. At a particular time of computation the value of a counting semaphore is 7. Then 20 P(wait) operations and 15 V(signal) operations were completed on this semaphore. What will be the resulting value of the semaphore?

IV. What is spinlock?

V. What are the two general hardware instructions that can be performed atomically?

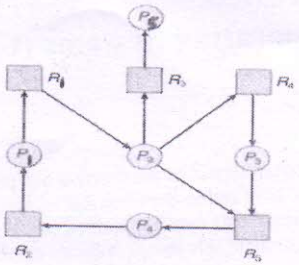
Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain Dekker's solution for critical section (CS) problem and prove that the algorithm satisfies all three requirements for the CS problem.

II. Define semaphore. What are the different types of semaphore? Provide the implementation of semaphore primitives.

III. Explain resource allocation graph(RAG) and construct the wait for graph for the following RAG-



IV. The structure of each process P_i, i = 0,1,2,3,……,9 is as follows:

```
while(1)
 {
    P(mutex)
    {CriticalSection}
    V(mutex)
 }
```

The code for P_{10} is identical except that it uses V(mutex) instead of P(mutex). What is the largest number of processes that can be inside the critical section at any moment (the mutex being initialized to 1)?

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. What are the four necessary conditions for characterizing deadlock? How we can prevent the occurrence of deadlock? Explain in detail.

II. State Reader –Writer synchronization problem and provide the solution of this problem using semaphores.

III. Consider the following snapshot of a system

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	2	0	0	1	4	2	1	2	3	3	2	1
P_1	3	1	2	1	5	2	5	2				
P_2	2	1	0	3	2	3	1	6				
P_3	1	3	1	2	1	4	2	4				
P_4	1	4	3	2	3	6	6	5				

Answer the following using the banker's algorithm-

a. Is the system in a safe state? Why?

b. If a request from process P1 arrives for (1,1,0,0), can the request be granted immediately.

IV. What is interprocess communication? Explain the two fundamental models of it with an example.

Mid Term Examination, 2016-17

Course: B.Tech. (CSE) **Year: II** **Semester: IV**

Object Oriented Programming (CSE4004)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. What will be the output of the program?

```
public class WrapTest
{
  public static void main(String [] args)  {
    int result = 0;
    short s = 42;
    Long x = new Long ("42");
    Long y = new Long (42);
    Short z = new Short ("42");
    Short x2 = new Short (s);
    Integer y2 = new Integer ("42");
    Integer z2 = new Integer (42);
    if (x == y)
      result = 1;
    if (x.equals(y) )
      result = result + 10;
    if (x.equals(z) )
      result = result + 100;
    if (x.equals(x2) )
      result = result + 1000;
    if (x.equals(z2) )
      result = result + 10000;
    System.out.println("result = " + result);  }
}
```

II. Do you need to include **Java.lang** explicitly? Show how to include all of the classes in a package called **pack**.

III. You have thread T1, T2 and T3, how will you ensure that thread T2 run after T1 and thread T3 run after T2?

IV. What is the default layout for an applet, frame and panel?

V. What will be the output of the program?

```
class Test
{
  public static void main(String[] args)
    {
     for(int i = 0; true; i++)
       {
       System.out.println("true");
       break;
       }
    }
}
```

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain the syntax for defining inheritance, final classes, super construct, and protected variables with examples.

II. Write an interactive program to compute the square root of a number. The input values must be tested for validity. If it is negative, the user-defined method MySqrt() should raise an exception.

III. Write a program that reads a statement from user input and prints how many words it has.

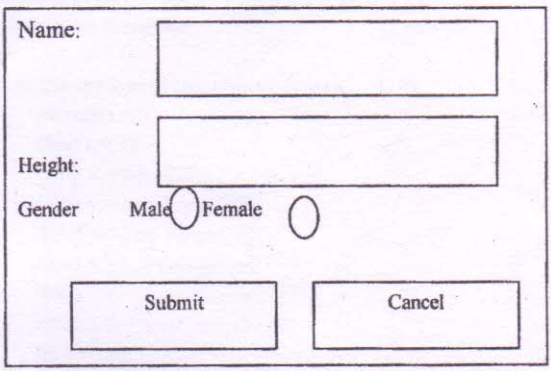
IV. Explain how a java GUI component handles the event with an example.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. An Australian university awards some grace marks to students who participate in the national games. Therefore, total marks awarded = Exam_Marks + Sports_Grace_Marks. If total marks scored are greater than maximum marks, then the final marks awarded will be equal to the maximum marks. An OO-based implementation will contain a class called Results, which extends a class called Exam, which itself extends a class called Student. It will also contain an interface called Sports, which is implemented by the Results class. The Results class will be responsible for computing the final marks scored by the students. Write a Java program.

II. Write a program in java to design the following window through AWT package.



III. Develop an applet that receives three numeric values as dynamic input from the user and then displays the largest of the three on the screen. Write a HTML page and test the applet.

IV. Compare and contrast:
(a) Java AWT and Swing.
(b) Abstract class and Concrete class
(c) Class and Interface

Mid Term Examination, 2016-17

Course: B.Tech.(CSE) **Year: II** **Semester: IV**

Computer Networks (CSE4003)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Frame type that refers to High-level Data Link Control error detection field is known as?

II. To guarantee detection of up to s errors in all cases, what is the minimum hamming distance in a block code?

III. What is the difference between CRC and Checksum?

IV. What is the maximum window size for data transmission using the selective reject protocol with n-bit frame sequence numbers?

V. What do you understand by address mapping?

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. What do you understand by sub netting? If the Network addresses: 192.168.10.0 and Subnet mask: 255.255.255.224 then how many subnet and host? And also write the valid subnet?

II. What is the Network ID, Broadcast Address, First Usable IP, or Last Usable IP on the sub network that the node 192.168.1.15/26 belongs to? subnet mask is 255.255.255.192.

III. Station A needs to send a message consisting of 9 packets to station B using a sliding window (window size 3) and

go back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no acks from B ever get lost) then what is the number of packets that A will transmit for sending the message to B?

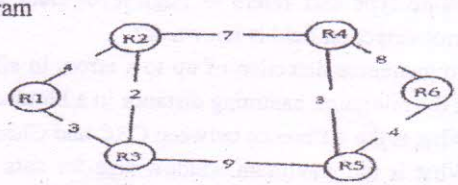
IV. What is the need of IP Addressing? And also differentiate IPv 4 and IPv6?

Section C

Note: Attempt any Three Questions (3x3=9)

I. Given the 8-bit data word 10110100, generate the 13-bit composite word for the Hamming code that corrects single errors and detects double errors.

II. Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram



All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbour with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

III. Why routing is needed? And also explain difference between the link state routing and distance vector routing?

IV. What is Stop-and-Wait Protocol? In stop and wait protocol , If the packet size is 1 KB and propagation time is 15ms the channel capacity is 109 bits / sec then find :

i) Transmission time

ii) Channel utilization

Mid Term Examination, 2016-17

B. Tech II Year IV Semester

Discrete Mathematics (CSE-4005)

Uni. Roll No: .....

Time: 90 minutes **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt ALL Questions. **(1x5=5)**

I. A die is rolled three times. Find the probability that exactly one odd number turns up among the three outcomes.

II. If p and q are false, is $(p \vee q) \wedge [(7\,p) \vee (7\,q)]$ true or false?

III. Consider the following:

p: This computer is good

q: This computer is cheap.

Write each of the following statement in Symbolic form.

a: This computer is not good but cheap.

b: This computer is costly but good.

IV. If T (x): x is teacher. M (x): is male, then write the Symbolic representation of the statement. " All teachers are male"

V. Write the principal conjunctive normal form of $p \wedge q$.

Section B

Note: Attempt any THREE Questions. (2x3=6)

I. A pair of fair dice is thrown. Find the probability p that the sum is 10 or greater if:
   a. 5 appears on the first die.
   b. 5 appears on at aleast one die.

II. The contrapositive of statement is given as :
" If $x < 2$ then $x + 4 < 6$"
Write the converse and inverse.

III. Prove that $(p \rightarrow 7\ q) \leftrightarrow (q \rightarrow p)$ is a contingency.

IV. Write down the negation of the following proposition for every number x there is a number y such that $y < x$.

Section C

Note: Attempt any THREE Questions. (3x3=9)

I. The contents of Urn 1, 2, 3, are as follows:

Urn 1: 1 white, 2 black, 3 red balls.
Urn 2: 2 white, 1 black, 1 red balls.
Urn 3: 4 white, 5 black, 3 red balls.

One Urn is chosen at random and two balls are drown they happen to be white and red. What is the probability that,

(a) They come from Urn 1 and
(b) They come from Urn 2 or 3.

II. Prove the validity without using truth Table of the following argument :

"If I get the job and work hard, then I will get promoted. If I get promoted then I will be happy. I will not be happy. Therefore, either I will not get the job or I will not work hard.

III. Show that: $\frac{1}{1.2} + \frac{1}{2.3} + - - + \frac{1}{n(n+1)} = \frac{n}{n+1}$

IV. Using the truth Table, find the principal disjunctive normal form and the principal conjunctive normal form of

$$(7\ p \vee 7\ q) \rightarrow (p \leftrightarrow 7\ q)$$

Mid-Term Examination

B.Tech II -Year, IV Semester, Session 2016-17

Introduction To Microprocessors (CSE-4002)

Time: 90 mins **Max Marks: 20**

SECTION-A

Note:Attempt All Questions

I. Why is instruction queue in 8086 microprocessor 6 byte long why not less than six or more than 6?

II. If the following program starts at 0100H, the contents of accumulator when PC reaches 0109H will be

```
LXI SP, 00FFH
LXI H, 0107H
MVI A, 20H
SUB M
```

III. Find the net time delays if the following set of instructions are executed. Assume the frequency of processor is 3 MHz.

```
    MVI C,FF H
BACK: DCR C
    JNZ BACK
```

IV. An 8085 program is given below. Assume that the carry flag is initially unset. The content of the accumulator after the execution of the program is…..

```
MVI A, 07H
RLC
MOV B, A
RLC
RLC
ADD B
RRC
```

V. What happen to the 8086 processor when it is reset?(write content of segment registers, flag register and instruction pointer register)

SECTION-B 2X3 =6 Marks

Note : Attempt any Three questions

I. How conditional flag is different from control flag in 8086? Draw the flag register of 8086 microprocessor.

II. Write a program to add two numbers of 4 digit BCD data stored in memory locations from 4200H to 4203 H and store the result from 4204H to 4206H

III. (a) What is the difference between minimum mode and maximum mode operation of 8086?

(b) Explain the concept of segmented memory, and also explain 20 bit physical address formation process.

IV. Explain following maximum mode signals of 8086

(a) $\overline{LOCK}$ (b) QS_0 and QS_1 (c) $\overline{s_0}$, $\overline{s_1}$ and $\overline{s_2}$

SECTION-C 3X 3=9 Marks

Note : Attempt any Three questions

I. Write an assembly language program to multiply two number of 16 bit data stored in memory location from 4200H to 4203H.store the product in memory locations from 4204H to 4207H.

II. Draw the pin diagram of 8086 microprocessor and explain the name of signals associated with Minimum mode operation of 8086.

III. Add two array having ten 8 bit numbers each and generate a third array of result. It is necessary to add the first element of array 1 with first element of array 2 and so on. The starting address of array 1,array 2 and array 3 are 2100H,2200H and 2300H respectively

IV. Explain the Different registers of 8086 microprocessor in detail?

Mid Term Examination
Even Semester, 2016-17

Course: B. Tech. II Yr. **Univ. Roll No.....**

Title of Paper: Environmental Studies **Paper Code: AHS-3001**

Time: 1.5 Hours **Max. Marks: 20**

Section- A

Note: Attempt all questions **1×5 = 5 Marks**

I. What is Eutrophication?

II. Give two examples of natural sources of pollution.

III. What is Greenhouse effect?

IV. Define 'Infant Mortality Rate'.

V. Give two advantages of Bio-Fuels.

Section-B

Note: Attempt any three questions **2×3 = 6 Marks**

I. 'Ecological impact of Acid rain is quite serious' comment.

II. Write short notes on solid waste disposal by:

(a) Land-Filling (b) Compositing

III. Discuss 'Hydrogen as an alternate future source of energy'.

IV. Discuss the effects of modern agriculture practices on environment.

Section-C

Note: Attempt any three questions **3×3 = 9 Marks**

I. Compare the arguments of Malthusians and Anti-Malthusians in present scenario.

II. Explain the term population explosion. Enumerate and discuss its effects.

III. Enumerate and discuss the sources, effects and control of Air pollution.

IV. Illustrate the photochemical reactions involved in the formation and destruction of Ozone layer in stratosphere. Also, briefly discuss the theme of 'Montreal protocol'.

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